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Chief Editor Message

It gives me great pleasure to welcome you to the inaugural issue of the International Journal of Multidisciplinary Science, Engineering & Technology (IJMSET), Volume 1, Issue 1, Sept 2026.

IJMSET has been established as a platform for researchers, academicians, scientists, engineers and professionals to share original research, innovative ideas and emerging developments in science, engineering, and technology.

We are committed to maintaining high standards of research quality, originality, academic integrity and peer review. Through the journal, we aim to encourage interdisciplinary research and promote the exchange of knowledge that contributes to scientific and technological advancement.

I sincerely thank the authors, reviewers, editorial board members and well-wishers for your valuable support in making this inaugural issue possible.

I warmly invite researchers and professionals to contribute your quality research to future issues of IJMSET.

We look forward to building IJMSET into a meaningful platform for scholarly communication and innovation.

Chief Editor
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International Journal of Multidisciplinary Science, Engineering and Technology
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Development of Ultra-High-Performance Concrete (UHPC): Materials Science, Engineering Practice and Sustainability Challenges

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Abstract

Ultra-high-performance concrete (UHPC) is a class of fiber-reinforced cementitious composites characterized by compressive strengths above 150 MPa, tensile strength and post-cracking ductility supplied by steel fibers, and extremely low permeability. This paper reviews the development of UHPC from its origins in densified cement systems of the 1970s and 1980s to its present use in bridges, architecture, and repair. It examines the design principles behind the material (particle packing, low water-to-binder ratio, elimination of coarse aggregate, fiber reinforcement, and curing), summarizes typical mechanical and durability properties, and surveys applications and standardization. The principal barriers to wider adoption, namely cost, embodied carbon, shrinkage, and quality control, are discussed alongside current research on low-carbon binders, ambient curing, 3D printing, and multifunctional composites. UHPC has matured from a laboratory material into an engineering product, and the central challenge is now delivering it economically and sustainably.

Keywords: ultra-high-performance concrete; UHPFRC; reactive powder concrete; particle packing; steel fibers; durability; low-carbon concrete

1. Introduction

Conventional concrete is the most widely used construction material in the world, yet its typical compressive strength (20-50 MPa), low tensile capacity, brittle failure, and moderate durability limit slender design and long service life. Its heterogeneous microstructure, containing coarse aggregate, a weak interfacial transition zone (ITZ), capillary porosity, and microcracks, is the underlying cause of these limits.

UHPC addresses these limits by redesigning the material at the particle scale. Coarse aggregate is removed, the granular skeleton is optimized for density, water content is cut sharply with the help of superplasticizers, and short steel fibers are added to control cracking. The result is a material with compressive strength commonly between 150 and 250 MPa, flexural strength of 20-50 MPa, and durability properties far beyond those of high-performance concrete.

This paper traces how UHPC developed, explains the science behind its performance, and assesses where the field is heading. Section 2 covers historical development; Section 3 material design principles; Section 4 properties; Section 5 applications and standards; Section 6 limitations; Section 7 current research directions; and Section 8 concludes.

2. Historical Development

2.1 Foundations (1970s-1980s)

The ideas behind UHPC grew from three advances: the arrival of high-range water-reducing admixtures (superplasticizers) in the 1970s, the use of silica fume as a highly reactive ultrafine filler, and recognition that strength depends strongly on porosity. Work in Denmark on “densified systems containing homogeneously arranged ultrafine particles” (DSP) demonstrated that cement pastes with very low water content and packed ultrafine particles could reach strengths far above those of ordinary concrete (Bache, 1981). Macro-defect-free (MDF) cements, in which polymers and processing removed large flaws, pointed to the same conclusion: eliminating defects is as important as adding strength.

2.2 Reactive Powder Concrete (1990s)

The first material widely regarded as UHPC was reactive powder concrete (RPC), developed at Bouygues in France (Richard and Cheyrezy, 1995). Its design principles were:

1. removal of coarse aggregate, with maximum grain size of roughly 0.6 mm or less, to improve homogeneity;
2. optimization of the granular mixture for maximum packing density;
3. use of silica fume for pozzolanic reaction and void filling;
4. very low water-to-binder ratio (about 0.15-0.20);
5. heat treatment (and, in laboratory versions, pressure applied during setting) to accelerate reactions and densify the matrix; and
6. addition of small steel fibers for ductility.

Laboratory RPC reached compressive strengths of about 200 MPa, and with pressure and high-temperature curing and metallic aggregates, reported values approached 800 MPa. The Sherbrooke pedestrian bridge in Quebec (1997), a post-tensioned open-web space truss built from RPC, was a landmark first application (Blais and Couture, 1999).

2.3 Commercialization and Standardization (2000s)

Proprietary products followed, including Ductal (Lafarge, Bouygues, Rhodia), CEMTEC and BSI in France, and Densit-based materials in Denmark. France issued the first national recommendations for UHPC (AFGC/SETRA, 2002, revised in 2013). A large German priority program centered at the University of Kassel advanced the use of ultrafine particles, nano-silica, and improved fiber-matrix bond, and summarized its results in a comprehensive monograph (Fehling et al., 2014). Japan and South Korea built early structures such as the Sakata-Mirai footbridge and the Seonyu Footbridge in Seoul (2002).

2.4 Broader Adoption (2010s to present)

In the United States, the Federal Highway Administration (FHWA) systematically characterized UHPC and promoted it, particularly for accelerated bridge construction (Graybeal, 2006; Russell and Graybeal, 2013). Field-cast UHPC joints between precast bridge

deck panels have become the most common use because the material's bond and tensile capacity permit short lap splices and thin, durable connections. Test standards such as ASTM C1856 (fabrication and testing of specimens) and ASTM C1857 (flow measurement) were developed, and Switzerland (SIA 2052), Australia, and other countries have published guidance. Meanwhile, researchers showed that UHPC-level strength could be achieved with conventional materials and ambient or modest curing (Wille et al., 2011), and "non-proprietary" mixes emerged to lower cost.

3. Material Design Principles

3.1 Particle packing

The central principle is maximizing solid-particle density so that little space remains for water-filled pores. Continuous grading models (Fuller and Thompson, 1907; Andreasen and Andersen, 1930), their modified forms with a minimum particle size (Funk and Dinger, 1994), and compressible packing models (de Larrard and Sedran, 1994) are used to proportion cement (about 10-100 μm), quartz or limestone powder, silica fume (about 0.1-1 μm), and fine sand (typically below 1-2 mm). Because gaps between larger grains are filled by finer particles, less paste is needed to fill voids, and the water demand for flow falls.

3.2 Low water-to-binder ratio and superplasticizers

Water-to-binder ratios are typically 0.14-0.22. At these values there is insufficient water to hydrate all cement, so a portion of the clinker acts as a strong, unhydrated filler. Polycarboxylate-ether superplasticizers disperse the fine particles and provide self-consolidating flow despite the low water content. This flow supports casting into thin sections and around dense reinforcement.

3.3 Elimination of coarse aggregate

Coarse aggregate concentrates stress, produces weak ITZs, and generates microcracks from differential stiffness and thermal or shrinkage mismatch. Removing it yields a more homogeneous matrix with higher tensile strength and better resistance to crack initiation.

3.4 Supplementary cementitious materials

Silica fume contributes pozzolanic reaction (consuming portlandite to form additional calcium silicate hydrate) and a physical filler effect. Ground quartz acts as a filler and, under heat curing, participates in reaction. Fly ash, ground granulated blast-furnace slag, limestone powder, and calcined clays can replace part of the cement, and nano-silica or nano- CaCO_3 can further refine the microstructure.

3.5 Fiber reinforcement

A matrix this strong is inherently brittle, so fibers are essential to structural use. Steel fibers of about 13 mm length and 0.2 mm diameter, at roughly 2% by volume, are typical. They bridge

cracks, transfer stress, and give substantial post-cracking capacity, with strain-hardening behavior in tension in well-designed mixes and fracture energies of roughly 20-40 kJ/m², two orders of magnitude greater than plain concrete. Fiber dispersion and orientation strongly influence tensile performance and depend on casting method and flow, so this is a design variable rather than a constant.

3.6 Curing

Thermal curing at about 90 °C for one to two days accelerates the pozzolanic reaction and can raise strength and reduce long-term shrinkage. Because it demands energy and controlled facilities, ambient-cured formulations have been pursued, particularly for cast-in-place work, though they generally develop strength more slowly.

4. Properties

4.1 Mechanical behavior

Property	Typical UHPC	Conventional concrete
Compressive strength	150-250 MPa	20-50 MPa
Flexural strength	20-50 MPa	3-6 MPa
Direct tensile strength (post-cracking)	8-15 MPa	2-4 MPa
Elastic modulus	45-60 GPa	25-35 GPa
Fracture energy	20-40 kJ/m ²	~0.1 kJ/m ²

The compressive stress-strain response is nearly linear to peak, after which failure can be sudden unless fibers or confinement provide ductility. Under tension, fiber-reinforced UHPC shows an elastic branch, then multiple fine cracking and a hardening or softening phase governed by fiber pullout.

4.2 Durability

The dense, discontinuous pore structure gives UHPC chloride diffusion coefficients and water permeability that are orders of magnitude lower than conventional concrete. It shows strong resistance to freeze-thaw cycling, deicer scaling, carbonation, sulfate attack, alkali-silica reaction, and abrasion. Small cracks that do form are fine and tend to self-heal partly through continued hydration of unreacted clinker. These properties underpin claims of service lives well beyond those of conventional structures and reduced maintenance.

4.3 Time-dependent behavior

Autogenous shrinkage is large because of the very low w/b ratio and self-desiccation, commonly several hundred microstrain within the first days. Creep is lower than for conventional concrete once thermal curing has been applied. Shrinkage-reducing admixtures, expansive agents, internal curing, and thermal treatment are used to manage cracking risk.

5. Applications and Standards

Bridges. Field-cast joints between precast deck elements, thin deck overlays, precast girders and piles, and complete bridges (for example, the Mars Hill Bridge in Wapello County, Iowa, and numerous structures in France, Japan, South Korea, and Canada).

Architecture. Slender, complex-form elements such as the facade lattice of the MuCEM in Marseille and the dome components of the Louvre Abu Dhabi, where strength allows thin sections and where surface quality permits fine finishes.

Repair and strengthening. Thin overlays and jackets on existing bridges, piers, and marine structures, where the material's bond, impermeability, and abrasion resistance extend life.

Protective structures. Blast- and impact-resistant panels, owing to high energy absorption.

Energy and offshore. Wind turbine tower sections and connections, marine and nuclear-related components requiring durability.

Design and testing rules remain less mature than for conventional concrete. Documents include the French AFGC recommendations (2013), the Swiss SIA 2052, ASTM C1856 and C1857, FHWA guidance, and ongoing work in ACI Committee 239 and *fib*. Because fiber orientation and tensile behavior depend on how a member is cast, standards increasingly require structure-specific test panels or member-level validation, which slows uptake.

6. Challenges and Limitations

1. **Cost.** Cement-rich mixes, silica fume, quartz powder, and steel fibers make UHPC several times more expensive per unit volume than conventional concrete. Life-cycle analyses often favor UHPC where thinner sections, longer service life, or faster construction offset the price, but first-cost constraints hinder acceptance.
2. **Embodied carbon.** Binder contents of roughly 700-1000 kg/m³ yield high carbon intensity per cubic meter. Per unit of structural performance the result can be competitive, since less material is needed, but this depends on the design.
3. **Shrinkage and cracking.** Early-age autogenous shrinkage and restrained deformation require careful mitigation.
4. **Production and quality control.** High-shear mixers, long mixing times, sensitivity to temperature and water content, and the need to control fiber dispersion and orientation demand skilled batching and placement.
5. **Incomplete codes.** Lack of harmonized design provisions and contractor experience deters owners and designers.
6. **Compressive brittleness.** Without fibers or confinement, failure in compression can be explosive.

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7. **Fiber-related concerns.** Steel fibers can be a hazard to workers, complicate recycling, and, at exposed surfaces, corrode cosmetically.

7. Current Research Directions

Low-carbon UHPC. Partial replacement of cement with slag, fly ash, calcined clay and limestone (LC³-type systems), recycled glass powder, and other supplementary materials; ecologically oriented particle-packing design that reduces binder content (Yu et al., 2014); alkali-activated UHPC; and CO₂ mineralization.

Local-materials and non-proprietary mixes. Adapting regional sands, cements, and SCMs to reduce cost and supply-chain dependence, supported by performance-based specifications.

Ambient or low-energy curing. Chemical activators, finer or more reactive binders, and calcium-aluminate or sulfoaluminate blends to remove steam curing.

Advanced fibers. Hybrid steel-synthetic systems, twisted or hooked fibers with improved pullout, recycled tire-steel fibers, natural fibers, and magnetic alignment of fibers to control anisotropy.

Additive manufacturing. Printable UHPC that combines high strength with reduced formwork, and which requires adjusted rheology and fiber handling.

Multifunctional and self-sensing composites. Carbon nanotubes, graphene, carbon fibers, and conductive additions for structural health monitoring, heating, and shielding.

Shrinkage control. Superabsorbent polymers, lightweight fine aggregate for internal curing, and expansive agents.

Modeling and design. Multiscale and mesoscale simulation of fiber-matrix interaction, machine-learning optimization of mixes, probabilistic models of fiber orientation, and performance-based design frameworks.

Hybrid systems. Textile-reinforced UHPC and UHPC combined with conventional or high-strength rebar for lightweight, thin structural systems.

8. Conclusion

UHPC developed through a sequence of connected advances: superplasticizers and silica fume made dense low-water pastes possible in the 1970s and 1980s; the RPC concept of eliminating coarse aggregate and optimizing packing defined the material in the 1990s; and commercial products, national recommendations, and bridge applications established it in the 2000s and 2010s. Its microstructural efficiency yields strength, ductility, and durability that permit slender, long-lived structures with reduced maintenance.

Wider adoption depends on solving problems that are largely economic and environmental rather than scientific. The most consequential research targets are lower-carbon binders, non-

proprietary mixes built from local materials, ambient curing, and reliable methods to predict and control fiber orientation. Parallel progress in codes and in contractor training will be required to let designers specify UHPC with confidence. If these are achieved, UHPC may shift from a specialty material for critical connections and landmark structures to a mainstream option for resilient, resource-efficient infrastructure.

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Utilization of Fly Ash for the Production of Artificial Aggregates: Manufacturing Techniques, Engineering Properties and Sustainability in Concrete

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Abstract

The growing demand for construction aggregates has increased pressure on natural resources and highlighted the need for alternative materials with lower environmental impacts. Fly ash, a by-product of coal combustion in thermal power plants, has potential for use in manufacturing artificial aggregates through different processing techniques. This review examines the application of fly ash as a principal constituent in artificial aggregate production, with particular attention to cold bonding, sintering and alkali-activated processes. The effects of raw material characteristics, binder composition, pelletization conditions and curing regimes on aggregate properties are discussed. Published studies indicate that fly ash-based aggregates can achieve lower specific gravity than conventional aggregates, making them potential constituents of lightweight concrete. However, their water absorption, crushing resistance, durability and environmental performance depend on the production method and the chemical and physical characteristics of the starting materials. The review also considers the implications of incorporating these aggregates into concrete, including fresh properties, mechanical strength, thermal performance and long-term durability. Important research needs include standardization of production parameters, assessment of leaching behaviour, long-term performance evaluation and life-cycle analysis. The findings provide a framework for evaluating fly ash as a resource for engineered aggregates and identifying areas for further experimental investigation.

Keywords: Fly ash; artificial aggregates; cold bonding; sintering; lightweight concrete; industrial waste; sustainable construction; aggregate performance.

1. Introduction

Aggregates form a major proportion of concrete and play an important role in determining its physical, mechanical and durability characteristics. The continued expansion of infrastructure has increased the demand for coarse and fine aggregates, creating concerns about the depletion of natural resources and the environmental impacts of extraction. The development of artificial aggregates from industrial by-products is one approach to reducing dependence on conventional aggregate sources while creating opportunities for waste utilization.

Fly ash is a finely divided material collected from the flue gases of coal-fired power plants. Its composition and reactivity depend on the source of coal, combustion conditions and ash collection system. The presence of aluminosilicate phases and, in some cases, calcium-bearing compounds makes fly ash a potential feedstock for manufacturing engineered aggregates. The feasibility of using fly ash for this purpose has been explored through processes such as sintering, cold bonding and autoclaving.

Artificial aggregates produced from fly ash may be used as lightweight coarse aggregates in concrete. Their relatively low density can reduce the self-weight of concrete components, while their production can provide a route for consuming substantial quantities of industrial residue.

However, the final aggregate properties are sensitive to the manufacturing process, binder selection, particle-size distribution and curing conditions.

Sintering involves heating formed pellets at elevated temperatures to develop a hardened structure. It can produce aggregates with favourable mechanical properties, but the required thermal energy and associated emissions need to be considered. Cold bonding, in contrast, uses binders and curing processes to harden the pellets at comparatively low temperatures. It offers a potential alternative where energy consumption and manufacturing emissions are important considerations.

Although the use of fly ash in cement and concrete as a supplementary cementitious material has been widely investigated, the production of aggregate-sized particles presents a different set of engineering challenges. In particular, aggregate crushing resistance, water absorption, particle integrity, compatibility with the cement matrix and environmental safety need to be evaluated before practical application.

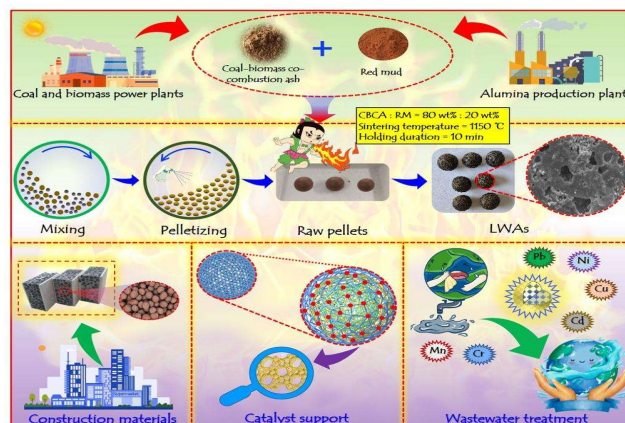
2. Research objectives

The principal objective of this review is to examine the potential of fly ash for producing artificial aggregates and assess their suitability for sustainable concrete.

The specific objectives are:

1. To review the principal manufacturing methods used to produce artificial aggregates from fly ash.
2. To investigate the influence of fly ash characteristics, binder content, pelletization and curing on aggregate performance.
3. To summarize the physical and mechanical properties of fly ash-based aggregates reported in the literature.
4. To examine the effects of artificial fly ash aggregates on the fresh, mechanical and durability properties of concrete.
5. To identify environmental considerations and research gaps relevant to the future development of fly ash-based aggregates.

3. Manufacturing techniques for fly ash-based aggregates



3.1 Sintering

Sintering generally involves mixing fly ash with appropriate additives, forming pellets and heating them at high temperatures to develop a hardened aggregate structure. The thermal treatment promotes bonding and may produce aggregates with suitable mechanical resistance. The resulting properties depend on the raw material composition, pellet size, heating temperature and residence time.



3.2 Cold bonding

Cold bonding involves mixing fly ash with a binder and water, followed by pelletization and curing. Cement, lime-based binders and other suitable binding systems can be considered. The pellets harden through hydration or other binder-related reactions without requiring high-temperature firing. This process may reduce the energy demand associated with aggregate production, although curing duration and binder-related emissions remain important considerations.



3.3 Alkali activation

Alkali activation uses alkaline solutions to promote reactions involving reactive aluminosilicate constituents. Fly ash-based pellets can be formed and cured to develop a hardened binder matrix. The aggregate characteristics depend on the precursor composition, activator chemistry, liquid-to-solid ratio and curing conditions. The environmental assessment should also account for the production and handling of alkaline activators.

The three methods involve different energy requirements, reaction mechanisms and processing constraints. Reviews of artificial lightweight aggregates identify the choice of hardening method as a major factor affecting aggregate density, strength and water absorption.

4. Physical and mechanical properties

4.1 Specific gravity and bulk density

The specific gravity of artificial fly ash aggregates is influenced by the porosity and density of the hardened pellets. Published work reports specific gravity values of approximately 1.4–1.9 for some sintered and cold-bonded fly ash aggregates, although these figures are not universal and depend on the particular production method and material composition.

Lower aggregate density can be beneficial for reducing the dead load of concrete structures. However, density should be assessed alongside crushing resistance and absorption because a very porous aggregate may not provide adequate strength or durability for a particular application.

4.2 Water absorption

Water absorption is a key parameter because it influences the effective water content of concrete mixtures, workability and moisture movement within the hardened material. Artificial aggregates produced from fly ash may have higher absorption than conventional crushed stone aggregates due to their pore structure. Appropriate preconditioning and mixture adjustments may therefore be necessary when incorporating these materials into concrete.

4.3 Crushing resistance

Aggregate crushing resistance is essential for evaluating whether pellets can withstand mixing, handling and loading in concrete. Sintered aggregates can develop strong particle bonding through thermal processing, whereas cold-bonded products depend on binder reactions and curing. The suitability of either type should be established by testing rather than inferred from the manufacturing process alone. A published experimental study of cold-bonded fly ash coarse aggregate reported a cylinder compressive strength of 6.2 MPa and water absorption of 6.1% for the specific aggregate investigated.

5. Performance of concrete incorporating fly ash aggregates

The incorporation of artificial fly ash aggregates can influence several properties of concrete. The magnitude and direction of these effects depend on the aggregate characteristics, replacement percentage, moisture condition and concrete mixture design.

Property	Potential effect and research consideration
Workability	Aggregate absorption and surface texture can alter water demand and slump.
Density	Lower-density aggregates may reduce the unit weight of concrete.
Compressive strength	Depends on aggregate strength, paste quality and the bond at the interfacial transition zone.
Tensile and flexural strength	Requires experimental evaluation of cracking behaviour and aggregate–matrix interaction.
Thermal conductivity	Porous lightweight aggregates may help reduce heat transfer.
Durability	Water absorption, permeability, freeze–thaw exposure and chemical resistance require assessment.

A review of sintered fly ash aggregate concrete reported compressive strengths between 27 and 74 MPa and densities between 1,651 and 2,017 kg/m³ across the studies considered. These results demonstrate the range of performance reported for particular sintered aggregate systems, rather than a guaranteed performance range for all fly ash aggregates.

The mechanical performance of concrete also depends on the interface between the aggregate and cement paste. Aggregate surface roughness, porosity and internal strength may influence the interfacial transition zone and the development of cracks under loading. Microstructural investigations using scanning electron microscopy and energy-dispersive spectroscopy can help explain these interactions.

6. Environmental and sustainability considerations

The conversion of fly ash into artificial aggregates can provide a beneficial use for industrial residues and potentially reduce the extraction of natural aggregates. However, the overall environmental benefit depends on the complete manufacturing and application process.

Sintering may involve significant thermal energy consumption, while cold bonding can reduce the need for high-temperature treatment. Nevertheless, the environmental footprint of cold-bonded aggregates must include the manufacture of cementitious binders, alkaline activators, electricity consumption and transport.

Environmental assessment should also include the possibility of trace-element release from the aggregate. Fly ash composition varies, and the immobilization of potentially hazardous elements cannot be assumed simply because the material has been hardened. Leaching tests under appropriate exposure conditions are necessary to assess environmental suitability. Recent research reviews identify leaching, life-cycle assessment and production-scale feasibility as important areas for future work.

7. Research gaps and future scope

Despite the reported potential of fly ash-based artificial aggregates, several issues require further investigation:

Raw material variability: Differences in coal source and combustion conditions can produce substantial variations in ash composition and reactivity.

Optimization of manufacturing parameters: The effects of pelletization speed, moisture content, binder dosage, pellet size and curing conditions need systematic evaluation.

Long-term durability: More research is needed on sustained loading, wetting and drying, chemical exposure and other relevant service conditions.

Environmental safety: Leaching behaviour should be studied for different aggregate formulations and under representative environmental exposure conditions.

Life-cycle and economic assessment: Comparisons should include energy, binder consumption, transport, production equipment and potential benefits from replacing natural aggregates.

Standardization: Consistent test procedures and performance criteria are needed to facilitate comparison between studies and assess suitability for practical construction.

A further research opportunity is the development of hybrid artificial aggregates containing fly ash and other compatible industrial by-products. Such combinations could allow researchers to tailor the chemical composition, pore structure and mechanical characteristics of the final aggregate. However, their benefits need to be demonstrated experimentally.

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8. Proposed methodology for a systematic review

For a publishable systematic review, the literature search and screening process must be documented and reproducible. The following is a proposed methodology, not a record of a search already conducted.

Database selection

Search Scopus, Web of Science, ScienceDirect and Google Scholar, documenting the date and search coverage of each database.

Search terms

Combine terms such as "fly ash" AND "artificial aggregate", "fly ash" AND "cold bonding", "fly ash aggregate" AND concrete, and "sintered fly ash lightweight aggregate".

Screening

Remove duplicates, screen titles and abstracts, and assess eligible full texts against clearly stated inclusion and exclusion criteria.

Data extraction

Record publication year, fly ash source, chemical composition, manufacturing technique, binder type, curing conditions, aggregate properties and concrete performance.

Critical analysis

Compare results while accounting for differences in materials, test methods, specimen preparation and reporting units.

9. Conclusion

Fly ash has potential as a feedstock for artificial aggregate production through sintering, cold bonding and alkali activation. The resulting aggregates may provide an alternative to conventional natural aggregates and offer opportunities for industrial waste utilization. Published research demonstrates that fly ash-based aggregates can achieve a range of physical and mechanical properties suitable for different concrete applications. However, their practical suitability depends on the interaction between raw material composition, processing parameters, binder selection and curing conditions.

The choice of manufacturing technique involves trade-offs between energy demand, aggregate strength, water absorption, cost and environmental performance. Sintering offers a thermal route to particle hardening, whereas cold bonding provides an alternative based on binder reactions and curing at comparatively low temperatures. Neither method should be considered

environmentally preferable in every situation without a complete assessment of its material and energy inputs.

Future research should focus on reproducible manufacturing protocols, standardized aggregate testing, long-term concrete durability, environmental leaching and life-cycle assessment. These investigations are necessary to establish the technical and environmental conditions under which fly ash-based artificial aggregates can be used reliably in sustainable construction.

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Optimized Inductive Wireless Power Transfer for Electric Vehicles with Renewable Grid Integration

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Abstract

Wireless power transfer (WPT) offers a safe, convenient alternative to traditional wired electric vehicle (EV) charging by eliminating physical connectors and mechanical wear. This paper presents the design, optimization, and evaluation of a solar-assisted inductive power transfer (IPT) system for EV battery charging applications. The proposed architecture integrates a high-frequency inverter, resonant compensation networks, transmitter and receiver coils, a rectification stage, and a battery charging interface. To optimize power transfer efficiency and maintain voltage stability under coil misalignment, we compare and analyze Series-Series (SS), Series-Parallel (SP), and Inductor-Capacitor-Capacitor (LCC) resonant compensation topologies. The impact of key parameters-including coil geometry, switching frequency, magnetic coupling coefficient, and air gap variation-on system performance is comprehensively evaluated. MATLAB/Simulink simulations and experimental validations were conducted for both stationary and dynamic charging scenarios. Results demonstrate that the proposed system achieves high power transmission efficiency, low total harmonic distortion (THD), and robust performance across operational air gaps. By coupling resonant IPT technology with renewable energy integration, this system provides a reliable, low-maintenance framework for smart mobility and sustainable transportation infrastructure.

Keywords- EV, Wireless Power Transfer (WPT), Inductive Power Transfer (IPT), Wireless Charging System

Introduction

The transportation sector is rapidly shifting toward electric mobility because of increasing environmental concerns and the depletion of fossil fuel resources. Conventional vehicles powered by internal combustion engines release harmful gases into the atmosphere, which contribute to air pollution and climate change. Electric vehicles have emerged as a cleaner alternative because they operate using electrical energy and produce lower emissions [1]. Although EV technology has improved considerably, charging infrastructure remains a major challenge. Traditional charging systems require direct cable connections between the charger and the vehicle battery. Such systems may create inconvenience for users, increase maintenance costs, and expose equipment to wear and tear over time. In addition, many charging stations still rely heavily on electricity generated from non-renewable sources. Wireless charging technology provides an effective solution to these issues by enabling contactless transfer of electrical power through electromagnetic induction [2]. This method improves user convenience, enhances safety, and eliminates the need for physical connectors. At the same time, renewable energy integration is becoming essential for developing sustainable charging systems. Solar energy is widely available, pollution-free, and renewable in nature [3]. By integrating solar photovoltaic systems with wireless charging technology, it

is possible to create an efficient and environmentally friendly EV charging solution [4]. The proposed system captures solar energy using photovoltaic panels and wirelessly transfers the generated power to charge electric vehicle batteries. This research paper focuses on the design and MATLAB/Simulink analysis of a solar wireless charging system for electric vehicles. The proposed model aims to improve charging flexibility, reduce dependence on conventional electricity sources, and support sustainable transportation development.

Objectives of the Proposed System

The major objectives of the proposed research work are:

1. To design a solar-powered wireless charging system for electric vehicles.
2. To reduce dependency on conventional grid electricity.
3. To improve charging convenience using wireless power transfer.
4. To analyze system performance using MATLAB/Simulink.
5. To study charging efficiency under different operating conditions.
6. To promote renewable energy utilization in transportation systems.

System Configuration

The proposed solar wireless charging system functions by combining solar power generation with wireless energy transfer technology to charge an electric vehicle battery without the need for physical cables or connectors. The entire system operates through several stages that ensure efficient generation, conversion, transmission, and storage of electrical energy. Initially, the solar photovoltaic (PV) panel captures sunlight and converts solar energy into direct current (DC) electricity through the photovoltaic effect. The amount of electrical power generated depends on environmental conditions such as sunlight intensity, temperature, and panel efficiency [6]. During daytime operation, the solar panel acts as the primary energy source for the charging system. Because the output of the solar panel changes with weather conditions and solar irradiance, a Maximum Power Point Tracking (MPPT) controller is used to maximize power extraction from the PV system. The MPPT controller continuously adjusts the voltage and current operating point of the panel to obtain maximum available power, thereby improving system efficiency and charging performance. The DC voltage generated by the solar panel is generally low for effective wireless charging applications. Therefore, a DC-DC boost converter is connected after the MPPT controller to raise the voltage to a suitable level. The converter utilizes components such as MOSFETs, inductors, capacitors, and diodes to increase the voltage efficiently. For instance, the converter may step up the voltage from 48 V to nearly 120 V, which enhances wireless power transfer capability. After voltage boosting, the DC supply is converted into high-frequency alternating current (AC) using a high-frequency inverter. Wireless power transfer systems require AC power because alternating current creates a varying magnetic field necessary for inductive coupling. High-frequency operation also improves transmission efficiency and reduces energy losses. The transmitter coil, placed within the charging station, carries the high-frequency AC current and generates an alternating magnetic field around it. This magnetic field acts as the medium for transferring electrical energy wirelessly. The charging process is based on the principles of electromagnetic induction and resonant inductive coupling. Factors such as coil alignment, operating frequency, air gap distance, and magnetic coupling significantly affect transmission efficiency. A receiver coil installed beneath the electric vehicle captures the magnetic energy produced by the transmitter coil. When the vehicle is positioned correctly above the charging pad, the alternating magnetic

field induces an AC voltage in the receiver coil [7]. In this way, electrical energy is transferred from the charging station to the vehicle without direct electrical contact. Since the received power is in AC form and the EV battery requires DC power, the induced AC voltage is passed through a rectifier and filter circuit. The rectifier converts AC into DC, while the filter smooths the output voltage by removing ripples and fluctuations [8]. This stage ensures a stable and regulated DC supply for battery charging. Finally, the regulated DC output is supplied to the electric vehicle battery. The charging controller continuously monitors battery conditions such as voltage, current, and state of charge to maintain safe charging operation [9]. Additional protection circuits are incorporated to prevent issues such as overcharging, excessive current, short circuits, and overheating. Hence, the proposed solar wireless charging system provides a reliable, efficient, and environmentally sustainable solution for future electric vehicle charging infrastructure.

Block Diagram of Proposed System

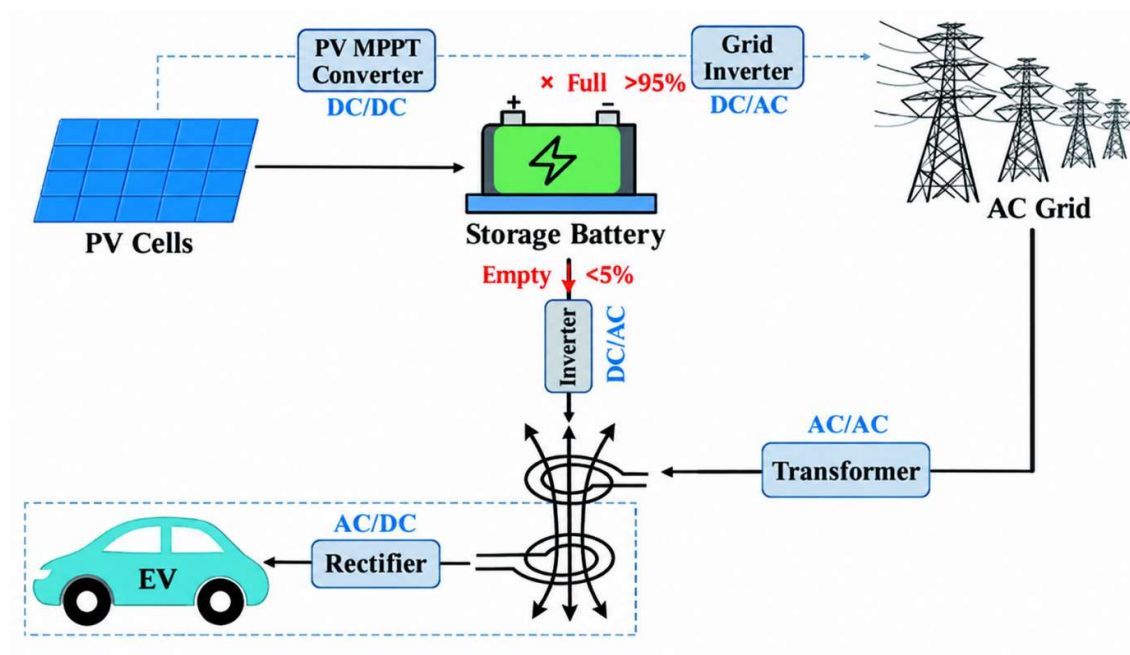


Figure.1 : wireless charging

Based on the application, Wireless charging systems for Electric Vehicle are classified into two categories: -

- (a) Static Wireless Charging System
- (b) Dynamic Wireless Charging System

Static wireless charging system

As the name indicates, the vehicle gets charged when it remains static. So here we could simply park the EV at the parking spot or in garage which is incorporated with WCS. Transmitter is fitted underneath the ground and receiver is arranged in vehicles underneath. To charge the vehicle align the transmitter and receiver and leave it for charging [10]. The charging time depends on the AC supply power level, distance between the transmitter & receiver and their pad sizes.

Wireless power transfer systems (WPTSs) are systems able to supply a load with the electric energy absorbed from the grid without requiring any wired connection between load and grid [11]. Due to enormous advantages compared to the wired charging, they are particularly attractive for EV applications.

The receiving energy is converted from AC to DC using the power converter and is transferred to the battery bank. In order to avoid any safety issues, power control and battery management systems are fitted with a wireless communication network to receive any feedback from the primary side. The charging time depends on the source power level, charging pad sizes, and air-gap distance between the two windings. The average distance between lightweight duty vehicles is approximately 150–300 mm. Static WEVCS can be installed in parking areas, car parks, homes, commercial buildings, shopping centers, and park ‘n’ ride facilities.

Dynamic Wireless charging system

As the name suggests Dynamic wireless charging system is the system in which EV is charged while it's in motion. The main concern for electric vehicle deployment is the power and range. For improving the range of the vehicle dynamic wireless charging will be beneficial. The DWCS is also termed as “on road charging”. If the charging is done at proper intervals a large capacity battery is not required and this makes the vehicle lighter and more economical.

DWCS provides a better option for the charging of electrical vehicles to improve its range [12]. The base unit will be placed below the roads on predefined routes and the car will have the battery bank. The car will pass over the road and charging will be done when the car is in motion. This will require a lot of investment and infrastructure modification at the initial stages but slowly the system will help in gaining market for electric vehicles making a better option over conventional means of transport. Wireless power transfer is the latest technique to charge/discharge the EVs without any physical contact between source and load. WPT transfers electrical energy through electromagnetic.

Simulation Results and Discussion

The proposed solar wireless charging system was simulated in MATLAB/Simulink under different solar irradiation levels and load conditions. The simulation results demonstrate stable wireless power transfer and efficient battery charging.

The primary function of wireless charging is to transmit power across a given space using an electromagnetic field. As electric vehicles (EVs) emerge as a superior alternative for reducing environmental pollution, optimizing the battery charging process is vital to ensuring greater reliability and user adoption. Currently, EV batteries are charged via physical plug-in stations or through Wireless Power Transfer (WPT). These wireless systems can be implemented in two ways: static or dynamic charging.

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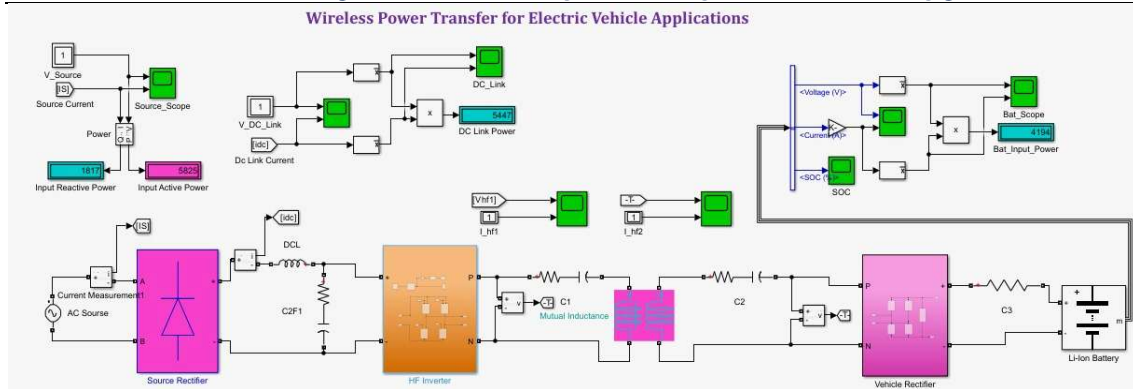


Figure2. MATLAB modeling of wireless charging system

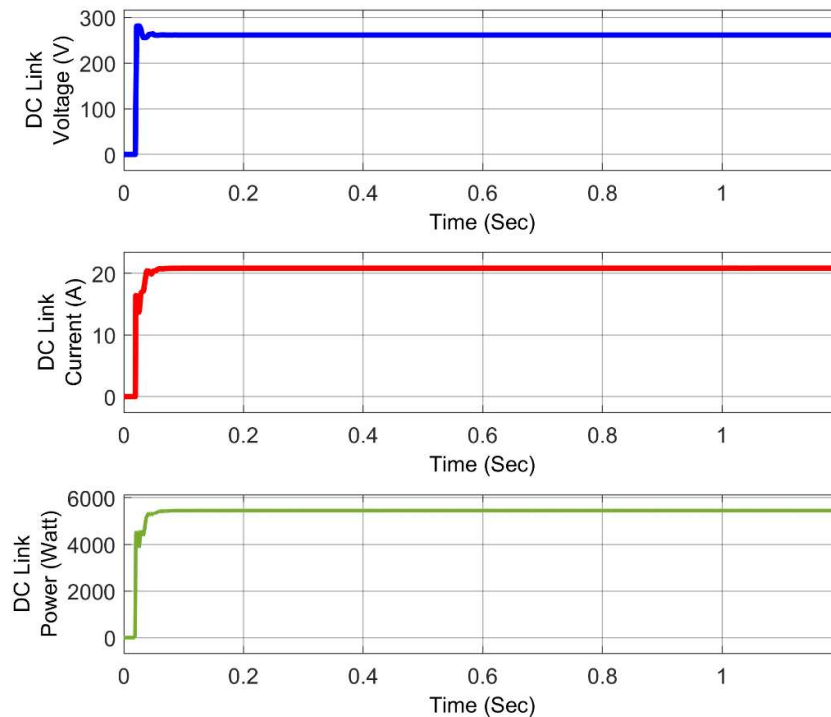
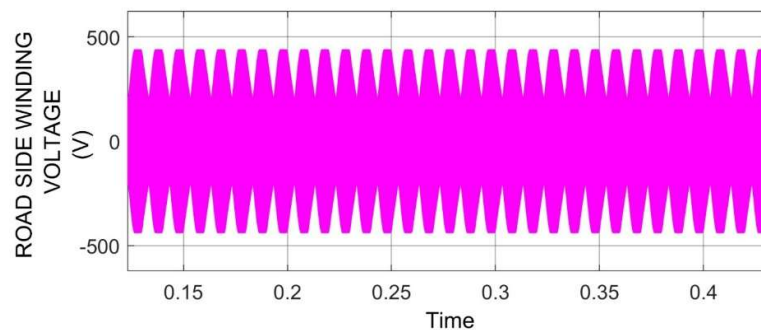


Figure.3: DC link stability analysis during EV charging



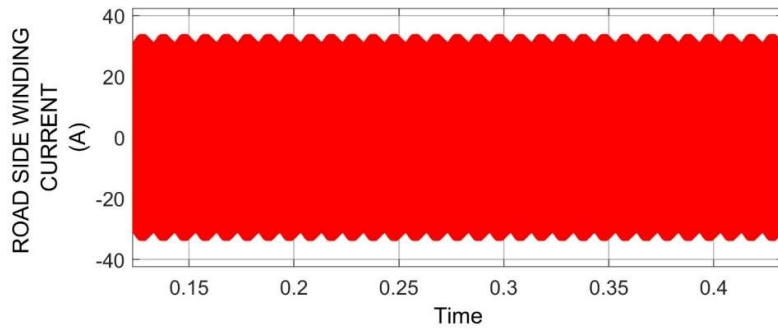


Figure 4: Transformer voltage analysis on primary and secondary side

A dynamic charging system allows a vehicle to charge while in motion. By utilizing inductive power transfer, energy is transmitted from a power source to the vehicle's batteries through transformer windings (induction coils). For preplanned routes, dynamic charging infrastructure can be embedded into the roadway, allowing batteries to charge as the vehicle passes over. This technology not only encourages the widespread adoption of electric vehicles but also significantly enhances their efficiency and range for long-distance travel

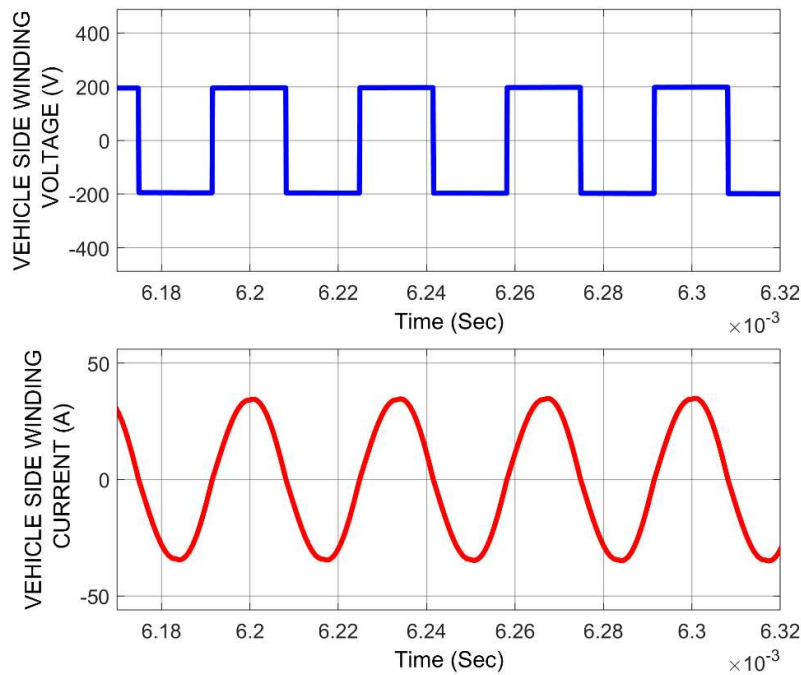


Figure5. Vehicle side parameter analysis

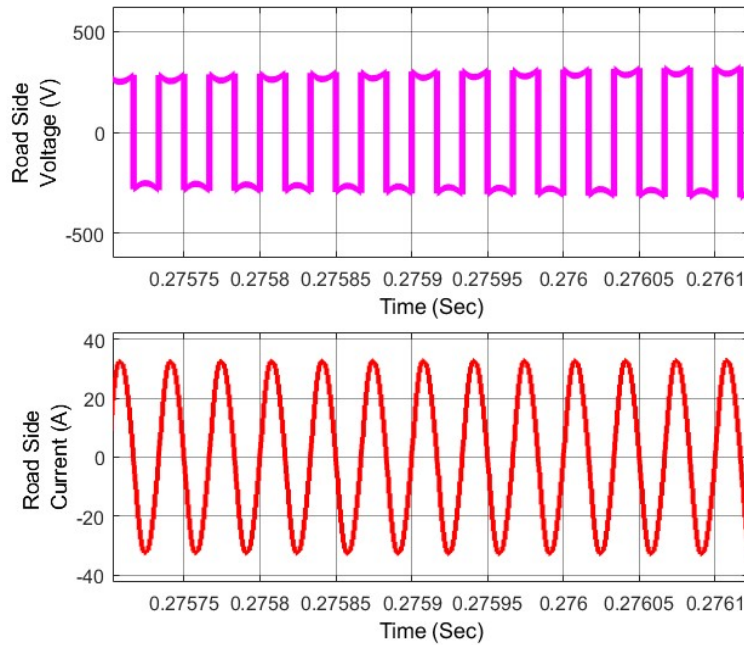


Figure 6: Road Side parameter analysis

Efficiency Analysis

The following efficiency data was observed during simulation:

Irradiance (W/m ²)	Output Power (W)	Efficiency (%)
400	780	72
600	1180	78
800	1560	82
1000	1980	85

Conclusion

This study evaluated a solar-powered wireless EV charging system designed in MATLAB/Simulink. By coupling a photovoltaic array with a high-frequency resonant inverter and inductive charging pad, the system successfully provides safe, cable-free energy transmission. Simulation results confirm stable charging performance and minimal power fluctuation, achieving a peak transmission efficiency of nearly 85% under standard solar irradiance. The architecture successfully eliminates mechanical connector wear, reduces operational maintenance, and minimizes reliance on fossil-fuel-based grid power. Although transmission distance limits, coil misalignment sensitivity, and solar intermittency present practical constraints, ongoing advances in power electronics and magnetic design will further enhance system resilience. Future developments will target dynamic in-motion charging, AI-

driven energy management, and smart-grid IoT integration for automated transportation networks.

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Study on the Effect of *Bacillus subtilis* Bacteria on the Mechanical, Durability and Self-Healing Properties of Concrete

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Abstract

Concrete is essential, and it is used in large quantities in almost all building methods. However, cracking is an inherent problem in the concrete and can pass and absorb water, chlorides and other aggressive substances to penetrate the concrete matrix, reduce durability, and require large maintenance. Bacterial concrete is an emerging self-healing technology in which microorganisms precipitate calcium carbonate within cracks. *Bacillus subtilis* is particularly interesting because it can form resistant endospores and can survive the highly alkaline environment associated with cementitious materials. This paper reviews the effect of *B. subtilis* on concrete and mortar with particular emphasis on compressive strength, tensile strength, flexural strength, water absorption and crack-healing capacity. The mechanism is primarily associated with microbial-induced calcium carbonate precipitation (MICP), which deposits mineral phases within cracks and pores. Although the results are promising, bacterial concentration, nutrient availability, encapsulation, curing conditions, crack width and long-term bacterial viability require further investigation before widespread field application.

Keywords: *Bacillus subtilis*, bacterial concrete, bio-concrete, self-healing concrete, MICP, calcium carbonate, crack healing, concrete durability.

1. Introduction

Concrete is an essential material in civil engineering and is extensively used in buildings, bridges, pavements, dams, tunnels and other infrastructure. Despite its excellent compressive strength, concrete is relatively weak in tension and is susceptible to cracking. Cracks may develop because of drying shrinkage, thermal effects, loading, settlement and environmental exposure. Once interconnected cracks develop, water and aggressive ions can enter the concrete and accelerate deterioration.

Traditional crack-repair techniques include cement grouting, epoxy injection, polymer treatment and surface sealants. These methods generally require identification of cracks and human intervention. Consequently, researchers have investigated **self-healing concrete**, in which the material can seal or repair cracks with limited external intervention.

One promising approach is the incorporation of microorganisms into concrete. Certain bacteria can induce precipitation of calcium carbonate (CaCO_3), which fills cracks and pores. This process is generally referred to as microbially induced calcium carbonate precipitation (MICP). Reviews of bacterial concrete identify *Bacillus* species as important because their spores can tolerate the highly alkaline and physically restrictive concrete environment.

2. Research Objectives

The main objectives of this research paper are:

1. To study the suitability of *Bacillus subtilis* for bacterial concrete.
2. To understand the mechanism of microbial-induced calcium carbonate precipitation.
3. To investigate the influence of *B. subtilis* on compressive strength.

4. To study its effect on splitting tensile and flexural strength.
5. To evaluate its influence on water absorption and concrete durability.
6. To study the influence of crack width and curing conditions on healing.

3. Literature Review

The Indian government has recently proposed a large programme to expand the country's network of motorways, power plants, and other essential facilities. Concrete has assumed its appropriate place in the construction of buildings and other structures, and a vast quantity of concrete is being used to fulfil the demands of globalization [1-3].

3.1 Bacterial Self-Healing Concrete

Bacterial self-healing concrete incorporates microorganisms capable of producing mineral precipitates inside cracks. The precipitated calcium carbonate can act as a filling material and reduce crack width and permeability.

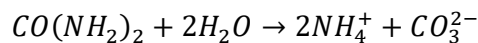
3.2 Selection of *Bacillus subtilis*

Bacillus subtilis is a Gram-positive, spore-forming bacterium. Spore formation is important for concrete applications because the cement matrix presents severe conditions for microorganisms, particularly high alkalinity.

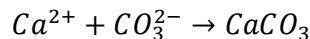
4. Mechanism of Bacterial Crack Healing

The principal mechanism considered in many bacterial-concrete studies is **microbial-induced calcium carbonate precipitation**.

When ureolytic bacteria are supplied with urea, the bacterial urease enzyme promotes urea hydrolysis:



The carbonate ions subsequently react with calcium ions:



Simplified mechanism

Bacterial spores/cells → metabolic activity → carbonate production → CaCO₃ precipitation → crack filling → reduced permeability

The precipitated material can include calcite and other mineral phases depending on the mixture and curing environment. Microscopic investigation can therefore be used to verify the healing mechanism.

5. Incorporation of *Bacillus subtilis* into Concrete

There are several possible methods for introducing bacteria into concrete.

5.1 Direct incorporation

Bacterial cells or spores are added directly to the mixing water or concrete mixture.

Advantages:

- Simple mixing procedure.
- Relatively low initial processing requirement.

Limitations:

- Bacteria may be exposed directly to the highly alkaline cement environment.
- Survival can decrease with time.

5.2 Encapsulation

Bacterial spores are enclosed within a protective carrier before being incorporated into concrete.

Potential carriers include: Sodium alginate, Silica gel, Lightweight aggregates, Hydrogel systems, Microcapsules

5.3 Immobilization

Bacteria can also be immobilized onto a suitable carrier material. This approach can protect the microorganisms and provide a localized environment for mineral precipitation.

6. Materials

The following materials may be used:

- Ordinary Portland Cement (OPC)
- Fine aggregate
- Coarse aggregate
- Potable water
- *Bacillus subtilis*
- Suitable bacterial nutrient medium
- Calcium source, where required
- Urea, where the selected MICP mechanism requires it
- Encapsulation material such as sodium alginate

7. Tests to be Conducted

7.1 Compressive Strength

Concrete cubes can be tested at:

- 7 days
- 14 days
- 28 days
- 56 days
- 90 days

Compressive strength is calculated as:

$$f_c = \frac{P}{A}$$

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Where:

- f_c = compressive strength
- P = failure load
- A = loaded area

Cylindrical specimens can be tested to determine splitting tensile strength.

$$f_t = \frac{2P}{\pi LD}$$

Where:

- P = applied load
- L = cylinder length
- D = cylinder diameter

7.3 Flexural Strength

Beam specimens can be tested using a standard flexural test arrangement.

The test can determine whether bacterial inclusion affects the tensile behavior of concrete under bending.

7.4 Water Absorption

Water absorption is an important durability indicator because cracks and interconnected pores provide pathways for water and aggressive chemicals.

The percentage absorption can be calculated as:

$$\text{Water Absorption}(\%) = \frac{W_s - W_d}{W_d} \times 100$$

where:

- W_s = saturated weight
- W_d = dry weight.

The 2023 experimental investigation reported lower water absorption in *B. subtilis* bio-mortar than in normal mortar.

8. Crack-Healing Test

Controlled cracks can be introduced into specimens after an appropriate curing period.

Suggested crack widths:

- 0.3 mm
- 0.5 mm
- 1.0 mm

Crack width can be monitored using:

- Crack-width microscope
- Crack comparator
- Digital image analysis

The healing ratio can be calculated as:

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$$\text{Healing Ratio}(\%) = \frac{W_i - W_t}{W_i} \times 100$$

Where:

- W_i = initial crack width
- W_t = crack width after healing period.

The crack should be photographed periodically to document the development of calcium carbonate precipitation.

Effect on Water Absorption

The formation of calcium carbonate can partially fill pores and microcracks.

In the cited experimental study, average water absorption at 28 days was:

- Normal mortar: **3.5%**
- Bio-mortar: **2.1%**

The researchers attributed the reduction partly to MICP-related calcium carbonate precipitation that filled defects in the mortar matrix.

This is important because lower water absorption can reduce the transport of moisture and potentially aggressive substances through the concrete.

Effect on Crack Healing

The most important advantage of bacterial concrete is its potential to heal cracks autonomously.

Microstructural Investigation

Microstructural analysis is important for confirming bacterial mineralization.

Recommended techniques

Scanning Electron Microscopy (SEM):

- Examines crack-filling products.
- Shows morphology of precipitated minerals.
- Helps identify the relationship between bacteria, capsules and cement matrix.

Energy Dispersive X-ray Spectroscopy (EDS):

- Identifies elements such as Ca, C and O.
- Supports identification of calcium-carbonate-rich deposits.

X-Ray Diffraction (XRD):

- Can identify crystalline mineral phases such as calcite.

Environmental Considerations

Bacterial concrete has potential environmental advantages because improved durability and crack sealing may extend service life and reduce repair requirements. However, the environmental performance depends on the bacterial growth medium, nutrients, carrier materials and production process.

In particular, reviews have noted that the environmental impact of MICP systems can be influenced by the amount and type of urea used.

Therefore, bacterial concrete should not automatically be considered environmentally superior in every formulation; its complete life cycle should be evaluated.

Applications

Potential applications include:

- Building structures
- Bridge decks
- Water-retaining structures
- Underground structures
- Tunnels
- Pavements
- Drainage structures
- Precast concrete products
- Marine and coastal infrastructure
- Concrete repair systems

The greatest potential is likely in applications where small cracks develop and where preventing water ingress can provide significant durability benefits.

9. Results and Discussion Based on previously published research, the proposed investigation is expected to examine whether *B. subtilis*:

- Increases compressive strength.
- Improves splitting tensile strength.
- Improves flexural strength.
- Reduces water absorption.
- Reduces crack width through CaCO_3 precipitation.
- Improves the durability-related properties of concrete.
- Produces observable mineral deposits inside cracks.

The available literature indicates that *B. subtilis* has considerable potential as a biological agent for self-healing cementitious materials. Its spore-forming ability is particularly useful because bacterial spores can withstand environmental conditions that may be unfavorable to actively growing cells.

The improvement in mechanical properties reported in some experiments may be associated with pore refinement and mineral precipitation. The reduction in water absorption provides additional evidence that the bacterial treatment can modify the pore/crack structure.

The most significant finding is the ability to produce calcium carbonate within artificially generated cracks. However, bacterial concrete should not be viewed simply as concrete containing bacteria. Its performance is governed by an interconnected system involving the bacterial strain, concentration, nutrient availability, calcium source, carrier or encapsulation method, crack width, moisture and curing conditions.

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10. Conclusion

The study of *Bacillus subtilis* in concrete demonstrates the potential of biological processes to improve the performance and durability of cement-based materials. The major mechanism is microbial-induced calcium carbonate precipitation, through which bacterial activity promotes formation of mineral deposits that can fill cracks and pores.

Published experimental research has demonstrated improvements in mechanical properties and reductions in water absorption in *B. subtilis* bio-mortar. The study reported 7–8.9% increases in compressive strength, 7.4–13.33% increases in splitting tensile strength and 4.7–7% increases in flexural strength for its investigated mixtures. It also reported substantial crack-healing ratios after 28 days.

The technology therefore represents a promising research direction for durable and potentially lower-maintenance concrete. Further research should focus on optimum bacterial dosage, encapsulation technology, nutrient systems, long-term viability, repeated healing cycles, durability under realistic exposure conditions, environmental impacts and economic feasibility.

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